

Limits – 8

Integer sequences: integrality, divisibility, periodicity

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Theoretical minimum.

Def. A sequence is integral if all of its terms are integers; the question “why is it integral” is almost always settled in one of three ways: a linear recurrence with integer coefficients; a conjugate pair $\alpha^n + \beta^n$; induction with a well-guessed auxiliary quantity.

Theorem 1 (a recurrence out of an irrationality). If $\alpha + \beta = p \in \mathbb{Z}$ and $\alpha\beta = q \in \mathbb{Z}$, then $s_n = \alpha^n + \beta^n$ are integers and $s_{n+1} = ps_n - qs_{n-1}$. Conversely, a relation of the form $a_{n+1} = ca_n + \sqrt{da_n^2 + e}$ usually hides such a pair: square it and subtract two consecutive equalities.

Theorem 2 (periodicity modulo m). Every integral sequence satisfying a linear recurrence of order k with integer coefficients and an invertible leading coefficient is periodic modulo every m (the Pisano period for Fibonacci numbers); hence its residues form a finite automaton.

Fact 1. Matrix form: $\begin{pmatrix} a_{n+1} \\ a_n \end{pmatrix} = A^n \begin{pmatrix} a_1 \\ a_0 \end{pmatrix}$, from which the gcd properties (for instance $\gcd(F_m, F_n) = F_{\gcd(m, n)}$) can be read off the powers of the matrix.

1. Prove Theorem 1 and apply it: (a) show that $\left(\frac{3+\sqrt{17}}{2}\right)^n + \left(\frac{3-\sqrt{17}}{2}\right)^n$ is an odd integer for every positive integer n (CIIM 2009); (b) find a recurrence for $\lfloor (2 + \sqrt{3})^n \rfloor$.
2. (Proposed problems) Let $a_1 = 2$ and $a_{n+1} = 2a_n + \sqrt{3(a_n^2 - 1)}$. Prove that all a_n are positive integers and that every odd prime p divides $a_p - 2$.
3. (Putnam 2018) Given a real number a , define $x_0 = 1$, $x_1 = x_2 = a$ and $x_{n+1} = 2x_nx_{n-1} - x_{n-2}$ for $n \geq 2$. Prove that if $x_n = 0$ for some n , then the sequence is periodic.

4. Prove that $\gcd(F_m, F_n) = F_{\gcd(m, n)}$ for Fibonacci numbers, and deduce that F_k divides F_n if and only if k divides n .
5. (IMC 2025 shortlist) A sequence of positive integers is defined by $a_1 = 1$, $a_2 = 1$, $a_3 = 2$ and $a_n a_{n-2} a_{n-3} = a_{n-1}^2 a_{n-3} + a_{n-1} a_{n-2}^2$ for $n \geq 4$. Prove that the sum of the reciprocals $S = \sum_{n \geq 1} \frac{1}{a_n}$ is irrational.